

OnlineBook Store

SQL Project Documentation & Database Analysis Queries

Project Overview

This project evaluates a relational database designed for an online retail book store. The backend system relies on three primary data tables, connected via standardized key identifiers that maintain consistent naming structures and data formats across files.

Dataset File	Primary Purpose	Key Linkage Field
Books.csv	Contains book details, pricing, inventory stock, and author metadata.	Book_ID
Customers.csv	Stores user profiles, geographical locations, and contact details.	Customer_ID
Orders.csv	Tracks transactional records linking buyers to purchasing history.	Book_ID , Customer_ID

1. Fundamental SQL Tasks (Basic Queries)

01. Extract all titles categorized under the "Fiction" genre.
02. Identify any literature published subsequent to the year 1950.
03. Display a complete roster of all patrons located in Canada.
04. Filter and display purchases that were finalized during November 2023.
05. Calculate the overall inventory count of books currently on hand.
06. Pinpoint the specific details of the highest-priced item in the catalog.
07. Identify any buyer who acquired more than one copy of a single title.
08. Extract all transaction records where the final billing amount is greater than \$20.

09. Provide a unique list of every genre represented within the primary Books dataset.

10. Locate the specific publication that currently holds the smallest remaining inventory.

11. Compute the aggregate financial revenue generated from all completed transactions.

2. Complex SQL Tasks (Advanced Queries)

01. Determine the total volume of literature sold, grouped by their respective genres.

02. Calculate the mean retail cost for publications specifically within the "Fantasy" category.

03. Identify patrons who have completed two or more separate purchasing transactions.

04. Discover which publication has the highest frequency of being ordered.

05. Isolate the top 3 most expensive titles that belong exclusively to the "Fantasy" genre.

06. Calculate the total volume of texts distributed, categorized by the author who wrote them.

07. Extract the geographical cities of buyers whose total spending exceeded the \$30 threshold.

08. Determine which individual patron invested the highest total amount of money into their purchases.

09. Compute the updated inventory levels that will remain after all current orders have been successfully processed and fulfilled.